理想 AI 眼镜 Livis
产品使用说明书

Li Auto Livis AI Glasses
User Manual

适用市场：美国（竞赛目标市场）
Intended market: United States (competition target market)

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适用产品 / Product: 理想 AI 眼镜 Livis / Li Auto Livis AI Glasses
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本说明书依据竞赛指定案例编写，不替代厂家正式说明。美国目标市场设定不代表已在当地上市、认证或建立服务网络。地区服务和未公开细节需向官方或购买渠道核实。

This competition-case manual does not replace official instructions. The U.S. target market does not establish local availability, certification, or service coverage. Confirm regional services and unpublished details with the manufacturer or seller.

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1. 关于本说明书 / About This Manual

1.1 说明书用途与适用范围 / Purpose and Scope

本说明书面向理想 AI 眼镜 Livis（以下简称“眼镜”）及配套充电盒的使用者，介绍产品功能、技术规格、首次设置、日常操作、安全、隐私、保养、故障排查和售后信息。请结合产品标签、正式随附资料和理想同学 App 中的说明使用本产品。

This manual is intended for users of Li Auto Livis AI Glasses (the “glasses”) and the charging case. It covers functions, specifications, setup, operation, safety, privacy, care, troubleshooting, and after-sales information. Use it together with the product label, official supplied documentation, and instructions in the Li Auto AI Assistant App (理想同学 App).

1.2 如何阅读本说明书 / How to Use This Manual

首次使用前先阅读第 5 章，再按第 6 章完成充电、连接和佩戴检查。日常功能见第 7 章；电池管理见第 8 章；拍摄和个人数据事项见第 9 章。清洁和更新见第 10 章，遇到异常时按第 11 章逐项排查，仍无法解决时按第 12 章申请支持。操作速查见附录 A。

Read Chapter 5 before first use, then follow Chapter 6 for charging, connection, and fit checks. See Chapter 7 for everyday functions, Chapter 8 for battery management, and Chapter 9 for photography and personal data. Chapter 10 covers care and updates. Use Chapter 11 to troubleshoot, and Chapter 12 to request support if the issue persists. Appendix A provides a quick reference.

1.3 警告、注意和提示的含义 / Warnings, Cautions, and Notes

警告：提示可能造成人身伤害的风险，以及应采取的避免措施。注意：提示可能导致设备损坏、数据丢失或操作失败的情况。提示：提供帮助理解功能或完成操作的补充信息。请先阅读相关提示，再执行步骤。图示中的箭头、编号和颜色仅用于辅助理解。

WARNING identifies a risk of personal injury and the action needed to avoid it. NOTICE identifies a risk of equipment damage, data loss, or an unsuccessful operation. NOTE provides additional information about a function or task. Read the relevant message before proceeding. Arrows, numbers, and colors in illustrations support the written instructions.

1.4 产品与软件版本说明 / Product and Software Versions

在标签、包装或 App 核对型号和软件。功能受配置、版本、账号、网络、语言、地区和服务影响。手册中计划上线的功能不等于已经可用，以实际更新与兼容说明为准。

Check model and software on the label, packaging, or app. Functions depend on configuration, version, account, network, language, region, and services. Planned features are not confirmed availability; check actual updates and compatibility.

2. 产品概述 / Product Overview

2.1 产品简介 / Product Introduction

Livis 将眼镜、第一视角相机、开放式音频和 AI 助手集成于可佩戴设备中。产品采用 Livis OS 系统，配合手机上的理想同学 App 使用。它通过按键、触控和语音完成交互，可用于记录日常场景、收听音频、通话及获取辅助信息。镜架与镜片配置不同，佩戴重量和光学效果也会不同。

Livis combines eyewear, a first-person camera, open-ear audio, and an AI assistant in a wearable device. It runs Livis OS and works with the Li Auto AI Assistant App on a phone. Buttons, touch controls, and voice commands support everyday

capture, audio, calls, and information assistance. Wearing weight and optical performance vary with the frame and lens configuration.

2.2 主要功能 / Main Features

功能 / Function	用途与条件 / Use and Conditions
第一视角影像 / First-person imaging	照片、Live Photo 及视频；保持视野清晰并遵守拍摄规定。 Photos, Live Photo, and video; keep the view clear and follow recording rules.
开放式音频与通话 / Open-ear audio and calls	音乐、手机通话及语音反馈；依赖适用的蓝牙连接与服务。 Music, phone calls, and spoken feedback; require the relevant Bluetooth connection and services.
录音与总结 / Recording and summaries	记录会议、学习等内容，支持转写、说话人区分及 AI 总结；先取得必要同意。 Record meetings or learning sessions with transcription, speaker identification, and AI summaries; obtain required consent first.
AI 问答 / AI assistance	语音问答、视觉问答、深度研究及记忆类功能；通常需要手机、网络及有效账号。 Voice and visual questions, deep research, and memory features; generally require a phone, network, and valid account.
车辆联动 / Vehicle integration	适用车型的控制、信息查询及音频联动；受车型、车机版本、账号授权和地区限制。 Controls, information, and audio integration for supported vehicles; depend on vehicle, software, account authorization, and region.
无线充电盒 / Wireless charging case	充电盒支持 USB-C 有线和 Qi 无线充电，眼镜放入盒中补能。 The case supports USB-C and Qi charging; place the glasses inside to recharge them.



图 2-1 日常使用流程概念示意：采集场景、语音辅助、手机查看。AI 生成，不代表实物结构或已验证的地区功能。
Figure 2-1. Conceptual workflow: capture, spoken assistance, and phone review. AI-generated; not a hardware or verified regional-function reference.

2.3 预期用途与使用限制 / Intended Use and Limitations

本产品用于日常佩戴、音频通信、个人影像记录及 AI 辅助信息获取，不作为工业防护眼镜、医疗设备或驾驶决策工具。AI 识别和回答可能出错，重要信息应独立核实。开放式音频可能被周围人员听见。传播手册没有说明眼镜配备显示屏；本文不将语音或手机反馈描述为镜片中的 AR 显示。

The product is intended for everyday wear, audio communication, personal capture, and AI-assisted information. It is not industrial protective eyewear, a medical device, or a driving decision tool. AI recognition and answers may be incorrect; verify important information independently. Nearby people may hear open-ear audio. The communication manual does not specify a display in the glasses; this manual does not present spoken or phone-based feedback as an AR lens display.

2.4 配套应用与服务 / Companion App and Services

配套应用为“理想同学 App”；英文名称是描述性译名，不代表美国商店正式名称。连接、媒体、设置和更新按当前界面操作。使用前确认美国下载、注册、语言和服务支持，音乐及车辆功能还受第三方服务和兼容性限制。

The app is 理想同学 App; the English name is descriptive, not a confirmed U.S. listing. Follow its current interface for connection, media, settings, and updates. Confirm U.S. download, registration, language, and services; music and vehicle functions also depend on third parties and compatibility.

3. 产品部件与包装内容 / Product Components and Package Contents

3.1 包装内容 / Package Contents

核对眼镜、随附充电盒及所购镜片。手册未列完整包装清单，USB-C 线、清洁布、袋、入门卡或适配器是否随附，以订单和实物清单为准。缺件或损坏时保留包装并联系卖家。

Check the glasses, included case, and ordered lenses. The source gives no complete packing list; check your order and package for a cable, cloth, pouch, guide, or adapter. Retain packaging and contact the seller for missing or damaged items.

3.2 眼镜部件示意图 / Glasses Overview

眼镜包含镜框和镜片、镜腿、硅胶鼻托、钛合金转轴，以及用于拍摄、收音和播放音频的部件。三个核心操作区域为触控板、记录按键和隐私提示灯。使用前辨认这些区域，避免遮挡镜头、提示灯和收音开孔。下图为传播手册原始产品尺寸图；标注尺寸不作为自行拆卸或改装依据。

The glasses include the frame and lenses, temples, silicone nose pads, titanium-alloy hinges, and components for capture, sound pickup, and audio playback. The three core control areas are the touchpad, recording button, and privacy indicator. Identify these areas before use and keep the camera, indicator, and microphone openings clear. The following original source diagram shows dimensions; it is not a disassembly or modification guide.



图 3-1 产品外观与尺寸图。来源：1.pdf 第 5 页；原图标注为毫米。

Figure 3-1. Product appearance and dimensions. Source: 1.pdf, page 5; original labels are in millimeters.

3.3 充电盒部件示意图 / Charging Case Overview

充电盒用于收纳和补能，支持 USB-C 及 Qi。按盒内形状自然放置，不挤压镜框或强行合盖。接口、接触结构和提示以实物为准；手册未确认独立配对键，不使用假定的盒上按键流程。

The case stores and recharges the glasses via USB-C or Qi. Fit the glasses naturally into the recess; do not squeeze the frame or force the lid. Check the actual ports, contacts, and indicators. No separate pairing key is confirmed; do not use an assumed case-button procedure.

3.4 按键、触控区域与状态提示 / Buttons, Touch Controls, and Status Indicators

操作区域 / Area	操作及结果 / Action and Result
触控板 / Touchpad	单击：播放/暂停、接听。双击：下一首；结束或拒接电话；退出 AI。三击：上一首。前滑/后滑：提高/降低音量。长按：唤醒理想同学。 Tap: play/pause or answer. Double-tap: next track, end/reject a call, or exit AI. Triple-tap: previous track. Swipe forward/backward: raise/lower volume. Press and hold: wake the assistant.
记录按键 / Recording button	单击：拍照。双击：开始/停止视频；录音时双击停止录音。长按：开始录音。 Press once: take a photo. Press twice: start/stop video; during audio recording, stop recording. Press and hold: start audio recording.
隐私提示灯 / Privacy indicator	录像时白色常亮；提示灯被遮挡时不能拍照或录像。手册未列出所有状态的灯色和闪烁含义。 Steady white during video recording; covering it prevents photos and videos. The source does not list all colors or blinking patterns.



图 3-2 触控板、记录按键及隐私提示灯。来源：2.pdf 第 13 页；编号 01—03 对应上表三个区域。

Figure 3-2. Touchpad, recording button, and privacy indicator. Source: 2.pdf, page 13; numbers 01–03 correspond to the three areas in the table above.

提示：同一操作在不同状态下有不同用途。先确认是否正在通话、播放或录音，再操作。长按时长和操作反馈以当前正式说明为准。

NOTE: The same gesture has different uses in different states. Check whether a call, playback, or recording is active before operating. Check the current official instructions for hold duration and feedback.

4. 技术规格 / Technical Specifications

4.1 尺寸、重量与材料 / Dimensions, Weight, and Materials

项目 / Item	规格 / Specification	来源 / Source
镜架重量 / Frame weight	36 g（约 1.27 oz），仅镜架，不含镜片 / frame only, excluding lenses	1.pdf p.4; 2.pdf p.2
镜架面宽间距 / Frame face-width spacing	134–137 mm（约 5.28–5.39 in）；定制热弯后约 134–136 mm / after custom heat bending, approximately 134–136 mm	1.pdf p.5
镜腿长度 / Temple length	169 mm（约 6.65 in）；尺寸图标 168.85 mm / drawing labels 168.85 mm	1.pdf p.5
镜架材料 / Frame material	TR90；钛合金转轴、硅胶鼻托 / TR90; titanium-alloy hinges and silicone nose pads	2.pdf pp.2–3
外观 / Finishes	亮光或哑光；经典黑、科技灰、橄榄绿 / glossy or matte; classic black, tech gray, olive green	1.pdf p.4; 2.pdf p.4

4.2 镜架与镜片选项 / Frame and Lens Options

镜片包括蔡司平光防蓝光、单色墨镜、感光变色及处方镜片；处方提供定制蔡司和 Superb 系列。度数、折射率和类型由专业人员按处方与订单确认。原设计针对中国脸型优化，购买前试戴，检查鼻托、耳部和镜腿压力。

ZEISS options include non-prescription blue-light-filtering, tinted, photochromic, and prescription lenses, with custom and Superb prescription series. Have a professional confirm prescription, index, and type. The original design targets Chinese facial profiles; try it on and check nose, ear, and temple pressure.

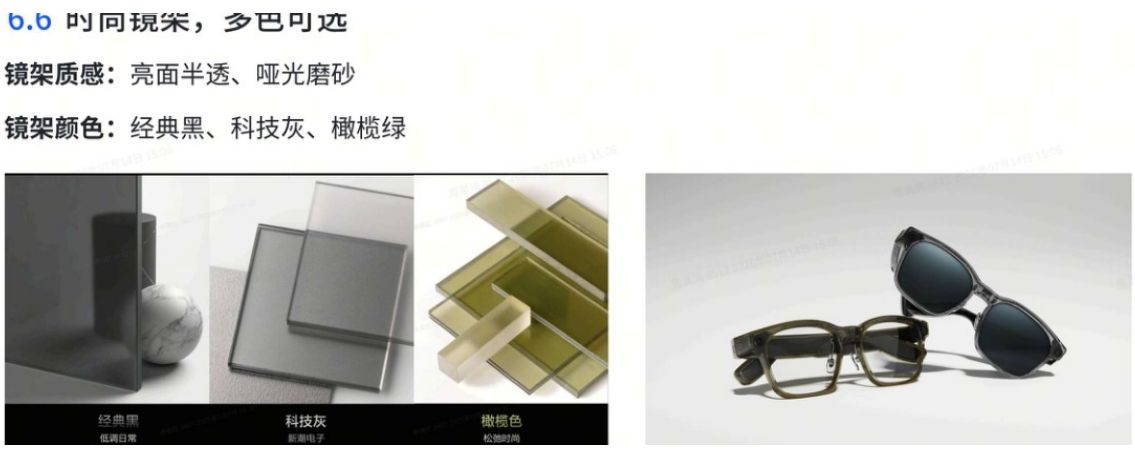


图 4-1 镜架配色与外观示例。来源：2.pdf 第 4 页；以购买配置为准。

Figure 4-1. Frame colors and appearance examples. Source: 2.pdf, page 4; your purchased configuration may differ.

4.3 相机与影像规格 / Camera and Imaging Specifications

项目 / Item	规格 / Specification	来源 / Source
相机 / Camera	1200 万像素，Sony IMX681 / 12 MP, Sony IMX681	1.pdf p.7
光学 / Optics	1G+4P；F/2.2；105°超广角 / 1G+4P; f/2.2; 105° ultra-wide angle	1.pdf p.7
照片 / Photos	4032 × 3024 像素 / pixels	1.pdf p.7
视频 / Video	1080p / 30 fps；单段最高 3 分钟 / up to 3 minutes per clip	1.pdf pp.7–8
防抖与动态照片 / Stabilization and Live Photo	EIS；Live Photo 记录约 3 秒动态内容，见实际版本 / EIS; approximately 3 seconds of motion, subject to version	1.pdf pp.7–8
画幅 / Aspect ratios	3:4、4:3、16:9、9:16 / 3:4, 4:3, 16:9, 9:16	1.pdf p.8

4.4 音频与无线连接 / Audio and Wireless Connectivity

项目 / Item	规格 / Specification	来源 / Source
音频 / Audio	开放式空间音频，双磁路扬声器 / open-ear spatial audio, dual-magnetic-circuit speakers	1.pdf pp.8–9
麦克风 / Microphones	360°全向录音，3 米内人声拾取描述 / 360° omnidirectional recording; source describes speech pickup within 3 m (9.84 ft)	1.pdf p.9

项目 / Item	规格 / Specification	来源 / Source
蓝牙 / Bluetooth	Bluetooth 5.4；最多同时连接 2 台设备，可保留 8 台设备绑定关系 / up to 2 simultaneous connections and 8 paired-device relationships	1.pdf p.10; 2.pdf p.1

拾音效果受噪声、距离和方向影响。手册未公布 Wi-Fi 版本及存储容量，本文不填入推测数值。

Noise, distance, and direction affect recording. The source does not specify a Wi-Fi version or storage capacity; no values are assumed.

4.5 电池与充电 / Battery and Charging

项目 / Item	手册数据 / Source Data	来源 / Source
眼镜电池 / Glasses battery	240 mAh；石墨负极电池 / graphite-anode battery	2.pdf p.4
典型续航 / Typical battery life	18.8 小时；实际受使用方式影响 / 18.8 h; varies with use	1.pdf p.5
眼镜充电 / Glasses charging	0–80%约 30 分钟；0–100%约 60 分钟 / approximately 30 min to 80%; 60 min to 100%	2.pdf p.4
充电盒电池 / Case battery	1700 mAh；约 4 次眼镜完整充电 / approximately 4 full glasses charges	1.pdf p.7; 2.pdf p.5
充电盒有线 / Wired case charging	USB-C；0–80%约 50 分钟；0–100%约 90 分钟 / approximately 50 min to 80%; 90 min to 100%	2.pdf p.4
充电盒无线 / Wireless case charging	Qi；最高支持 15 W；0–80%约 2 小时 15 分，0–100%约 3 小时 / up to 15 W; approximately 2 h 15 min to 80%, 3 h to 100%	2.pdf p.5
视频录制总时长 / Total video recording time	手册标示 41 分钟，不等于单段时长 / stated as 41 min, not the per-clip limit	1.pdf p.8

4.6 防护等级与使用环境 / Ingress Protection and Operating Conditions

眼镜防护等级为 IP54：可提供一定防尘和防溅水保护，不代表完全防尘或可浸水。请勿游泳、淋浴或用水冲洗电子镜架。传播手册未给出充电盒防护等级，也未给出工作、充电及存储温度范围；保持充电盒干燥，避免高温、凝露和过冷环境，具体温度限制按正式标签或说明核实。

The glasses are rated IP54, providing limited dust and splash protection rather than complete dust exclusion or immersion protection. Do not swim, shower, or rinse the electronic frame. The source does not state a case ingress-protection rating or operating, charging, and storage temperature ranges. Keep the case dry, avoid excessive heat, condensation, and very cold conditions, and confirm numerical limits from official labeling or instructions.

4.7 性能数据的测试条件 / Performance Test Conditions

充电和续航数据为手册典型值，不是用户保证。手册充电时间有差异，本稿采用 2.pdf 第 4—5 页专门章节：充电盒有线满充约 90 分钟。18.8 小时续航、41 分钟总录像和充电盒补能是不同指标，不应相加。

Charging and battery-life figures are typical, not guaranteed. Source sections differ; this manual follows the dedicated battery section in 2.pdf, pages 4–5, including approximately 90 minutes for a full wired case charge. The 18.8-hour use, 41-minute total video, and case recharge measures must not be added.

手册列 300 ms 唤醒、800 ms 普通任务响应和 715 ms 抓拍实验室数据；任务、网络和测试方法会改变结果，不能视为固定等待时间。换算：1 in = 25.4 mm，1 oz ≈ 28.3495 g。

Laboratory figures include 300 ms wake-up, 800 ms task response, and 715 ms capture. Task, network, and method affect results; these are not fixed waiting times. Conversions: 1 in = 25.4 mm; 1 oz ≈ 28.3495 g.

5. 安全警告 / Safety Warnings

本章包含眼镜和充电盒的重要安全信息。请在首次充电或佩戴前阅读，并在需要时提供给其他使用者。

This chapter contains important safety information for the glasses and case. Read it before charging or wearing them for the first time, and make it available to other users.

5.1 一般安全注意事项 / General Safety Precautions

警告：请勿自行拆解、改装、刺穿或维修眼镜和充电盒。内部电池和电路损坏可能造成短路、发热或伤害。请避免跌落、重压和强烈撞击。发现结构破损或电路外露时停止使用并联系支持。维修对保修的影响应按适用条款及法律认定，不以本说明书替代保修政策。

WARNING: Do not disassemble, modify, puncture, or repair the glasses or case yourself. Damage to internal batteries and circuits may cause short circuits, heat, or injury. Avoid drops, pressure, and strong impacts. Stop use and contact support if the structure is damaged or circuits are exposed. Any warranty effect of repair is determined by applicable terms and law; this manual does not replace the warranty policy.

5.2 佩戴与镜片使用安全 / Wearing and Lens Safety

更换处方镜片或调节镜架时，请联系产品支持渠道或具备相应资质的配镜人员，不要自行撬动镜片、热弯镜架或拆开镜腿。本产品不是工业防冲击眼镜，不用于化学喷溅、打磨、强激光等危险作业。若佩戴部位红肿、发痒或出现眩晕，立即停止佩戴；症状持续时咨询医生。

For prescription-lens replacement or frame adjustment, contact product support or a suitably qualified optical professional. Do not pry out lenses, heat-bend the frame, or open the temples yourself. The product is not industrial impact-protection eyewear and must not be used for chemical splashes, grinding, or strong laser exposure. Stop wearing it if you develop redness, itching, or dizziness; seek medical advice if symptoms persist.

5.3 听力保护 / Hearing Protection

警告：长时间高音量听音可能损伤听力。开始播放前调低音量，再缓慢提高到舒适水平。开放式音频不意味着高音量安全；不要在嘈杂环境中持续提高音量以掩盖噪声。出现耳鸣、听觉钝化或不适时暂停使用。公共场所应控制音量并尊重周围人员。

WARNING: Prolonged high-volume listening may damage hearing. Start at a low volume and increase gradually to a comfortable level. Open-ear audio does not make high-volume use safe; do not keep raising it to mask noise. Pause use if you notice ringing, reduced hearing, or discomfort. Keep volume considerate in public places.

5.4 驾驶与注意力安全 / Driving and Situational Awareness

步行、骑行、驾驶和操作机械时保持对周围环境的注意，遵守所在地法规。不要因拍摄、查看手机、语音问答或设置设备分心。需要复杂操作时先到安全地点停车或停止活动。车辆联动功能不能代替驾驶员判断或车辆正式操作要求；确认指令结果时以车辆实际状态为准。

Stay aware while walking, cycling, driving, or operating machinery, and follow local law. Do not let capture, phone viewing, voice questions, or settings distract you. Stop safely before performing complex tasks. Vehicle integration does not replace the driver's judgment or official vehicle instructions; confirm command results against the actual vehicle state.

5.5 电池与充电安全 / Battery and Charging Safety

警告：使用与产品标示输入规格相符、状态良好的充电设备。不要使用破损线缆，不要在潮湿接口、被覆盖或通风不良的环境中充电。发现明显异常发热、冒烟、异味、鼓包或变形时，若能安全操作则断开外部电源，停止使用并联系支持；不要触碰泄漏物或继续试充。危险无法控制时远离设备并联系当地紧急服务。

WARNING: Use undamaged charging equipment that matches the labeled input requirements. Do not use damaged cables or charge through wet ports, under coverings, or in poorly ventilated conditions. If you notice unusual heat, smoke, odor, swelling, or deformation, disconnect external power if safe, stop use, and contact support. Do not touch leaking material or try charging again. Move away and contact local emergency services if the hazard cannot be controlled.

5.6 防水、防尘与温度注意事项 / Water, Dust, and Temperature Precautions

IP54 不表示眼镜完全防水或完全防尘。避免浸水、淋浴、水流冲洗、大量汗液积聚或粉尘作业；接触水后用软布擦干，完全干燥后再放入充电盒。充电盒防护等级未在传播手册中公布，应保持干燥。不要放在暴晒的车内、热源旁或可能凝露的地方；温度明显不适宜时移至室内温和环境，待设备恢复后再使用或充电。

IP54 does not mean waterproof or completely dustproof. Avoid immersion, showering, running-water rinsing, heavy sweat accumulation, and dusty work. Wipe moisture away with a soft cloth and let the glasses dry fully before placing them in the case. The source does not publish a case protection rating; keep it dry. Avoid sun-heated vehicles, heat sources, and condensation. If conditions are unsuitable, move indoors and let the device return to a moderate temperature before use or charging.

5.7 儿童接触与小部件风险 / Children and Small Parts

警告：将产品、附件及包装中的小部件置于幼儿和宠物无法触及的位置，避免误吞、缠绕或损坏电池。儿童使用时由成年人监督，并确认佩戴尺寸适合、音量适中、已理解隐私与安全规则。传播手册未给出最低使用年龄，本说明书不自行设定厂家年龄限制。

WARNING: Keep the product, accessories, and small packaging parts away from young children and pets to prevent swallowing, entanglement, or battery damage. Supervise children, ensure a suitable fit and volume, and explain privacy and safety rules. The communication manual gives no minimum age; this manual does not invent a manufacturer age limit.

5.8 产品损坏及异常处理 / Damage and Abnormal Operation

镜片破裂、镜腿开裂、异常发热或异味、明显进液、充电接触部位腐蚀时，应停止佩戴和充电。不要用胶水粘合、金属工具清理内部或通过反复开机验证故障。保留故障发生时间和现象，按第 12 章联系支持。只有普通连接或设置问题，才按第 11 章进行常规排查。

Stop wearing and charging the product if lenses or temples are broken, unusual heat or odor occurs, liquid has entered, or charging contacts are corroded. Do not glue parts together, clean internally with metal tools, or repeatedly power it to test a fault. Record when and how the issue occurred and request support under Chapter 12. Routine troubleshooting in Chapter 11 is for ordinary connection or settings issues only.

6. 首次使用 / Getting Started

6.1 使用前准备 / Before You Begin

检查产品无损坏、镜片配置正确，阅读第 5 章。准备兼容手机、账号和稳定网络，按官方下载要求确认系统版本及美国地区可用性；手册未公布最低手机系统版本。

Check for damage and the correct lenses, then read Chapter 5. Prepare a compatible phone, account, and stable network. Check official system requirements and U.S. availability; minimum phone OS versions are not specified in the source.

6.2 首次充电 / Initial Charging

1. 检查眼镜和充电盒接触部位干燥、无异物。

1. Check that contact areas on the glasses and case are dry and clean.

2. 自然折叠镜腿，将眼镜按盒内形状放入，确认稳妥就位；不要强行压入。

2. Fold the temples naturally and place the glasses into the shaped recess. Make sure they sit securely; do not force them.

3. 用适用 USB-C 线和符合输入要求的电源为充电盒充电。状态显示以实物或 App 为准，不按未经核实的灯色判断。

3. Charge the case with a suitable USB-C cable and power source matching its input requirements. Check the actual device or app status rather than assuming unverified indicator colors.

提示：首次使用前确保电量充足，不需按固定时间强制充电。异常发热时停止充电，按第 5 章处理。

NOTE: Ensure sufficient power for first use; there is no fixed mandatory first-charge duration. Stop charging for unusual heat and follow Chapter 5.

6.3 安装与设置配套应用 / Installing and Setting Up the Companion App

1. 通过理想官方提供的下载入口获取“理想同学 App”，核对开发者信息；不要使用来源不明的安装包。

1. Obtain 理想同学 App through a download route provided by Li Auto and check the developer. Do not use an unknown installation package.

2. 阅读服务条款和隐私政策，按界面完成账号登录或注册。

2. Read the terms and privacy policy, then sign in or register as prompted.

3. 进入 App 的眼镜专区或设备管理入口，准备添加设备。名称随版本可能不同。

3. Open the glasses section or device-management entry to add the device. Labels may vary by version.

6.4 连接与配对 / Connecting and Pairing

1. 开启手机蓝牙，确保眼镜有电并靠近手机。

1. Enable phone Bluetooth and keep the powered glasses close to the phone.

2. 从 App 的添加设备入口选择眼镜，按当前提示使设备进入可发现或配对状态。

2. Choose the glasses under Add Device in the app and follow its current instructions to make them discoverable or ready to pair.

3. 核对设备名称并确认配对；出现手机系统蓝牙请求时按提示确认。

3. Check the device name and confirm pairing, including any phone-system Bluetooth request.

4. 等待 App 显示已连接，再做一次音频或拍照测试。若只显示系统蓝牙连接而 App 未识别，按第 11.1 节排查。

4. Wait for the app to show a connection, then test audio or a photo. If Bluetooth connects but the app does not recognize the glasses, use Section 11.1.

提示：首次配对流程以 App 当前指引为准。

NOTE: Follow current app guidance for first pairing.

6.5 必要权限与账号设置 / Required Permissions and Account Settings

按系统和所需功能开启权限：蓝牙或附近设备用于连接；相册或存储用于导入保存媒体；麦克风权限在 App 需要手机收音时使用；通知用于接收相关提醒；位置权限仅在确有相关功能需求时开启。网络用于在线 AI、服务和更新。不要将手机相机权限与眼镜自带相机混为一谈。拒绝非必要权限后，可保留其他不依赖该权限的功能。

Grant permissions according to the system and functions you use: Bluetooth or Nearby Devices for connection; Photos or storage for saving media; microphone access when the app needs phone audio; notifications for alerts; location only when a relevant feature requires it. Online AI, services, and updates need network access. Phone-camera permission is distinct from the camera in the glasses. Denying an optional permission need not disable unrelated functions.

6.6 佩戴与调节 / Wearing and Adjustment

1. 双手拿住镜腿，轻放于鼻梁和耳部，检查镜框稳定且无明显压迫。

1. Hold both temples and place the glasses gently on the nose and ears. Check that the frame is stable without uncomfortable pressure.

2. 保持镜片清晰、镜头和提示灯无遮挡；如鼻托或镜架需要明显调节，联系专业人员。

2. Keep the lenses clear and the camera and indicator unobstructed. Contact a professional if substantial nose-pad or frame adjustment is needed.

3. 试拍后在手机上查看画面，确认视角合适；不要假设镜片中有取景显示。

3. Take a test photo and review it on the phone to check the view. Do not assume there is a viewfinder in the lenses.



图 6-1 佩戴姿态示意，沿用原稿图；非尺寸依据。

Figure 6-1. Illustrative posture retained from the draft; not a dimensional reference.

6.7 首次使用检查 / First-Use Checklist

检查 / Check	完成标准 / Completion
连接 / Connection	App 识别眼镜且连接正常 / App recognizes and connects to the glasses
佩戴 / Fit	稳妥舒适，镜头与隐私灯无遮挡 / Comfortable and stable; camera and indicator clear
音频 / Audio	低音量播放可听见；单击可暂停 / Audible at low volume; tap pauses playback
拍摄 / Capture	单击记录键后能在 App 找到测试照片 / Test photo appears in the app after one button press
AI / Assistant	说“理想同学”或长按触控板后获得响应，网络和服务可用 / Assistant responds to 理想同学 or a touchpad hold when network and service are available
隐私 / Privacy	确认测试环境允许拍摄或录音 / Recording is permitted in the test setting

7. 日常操作 / Everyday Operation

7.1 按键与触控操作速查 / Button and Touch Control Reference

按键操作与触控操作需分别理解：记录按键用于拍照、视频和录音，触控板用于播放、音量、通话和 AI。完整速查见附录 A。不要把触控板双击与记录按键双击混为一谈。

Distinguish the recording button from the touchpad: the button controls photos, video, and audio recording; the touchpad controls playback, volume, calls, and AI. See Appendix A for the full reference. A touchpad double-tap is not the same as two recording-button presses.

7.2 拍摄照片 / Taking Photos

使用前提：眼镜有电，镜头和隐私提示灯无遮挡，场所允许拍摄。保持头部平稳，单击记录按键拍照；完成后在 App 媒体管理页面查找照片。拍摄敏感资料或他人时先取得必要许可。模糊照片可通过改善光线、清洁镜头和保持稳定后重试。

Before use: ensure sufficient power, a clear camera and privacy indicator, and permission to photograph. Hold your head steady and press the recording button once. Review the photo in the app’s media section. Obtain required permission before capturing people or sensitive material. If an image is blurred, improve lighting, clean the camera, and retry while steady.

7.3 录制视频 / Recording Videos

使用前提：电量及可用存储充足，提示灯无遮挡。双击记录按键开始录像，再双击停止；录像时隐私灯白色常亮。手册标示单段最高 3 分钟，软件版本可能影响可选时长。结束后检查 App 中的文件是否完整，不把总计 41 分钟的实验描述理解为单段录像上限。

Before use: ensure adequate power and storage and an unobstructed privacy indicator. Press the recording button twice to start video and twice again to stop. The privacy indicator stays white during recording. The source states up to three minutes per clip; available durations may vary with software. Check that the file is complete in the app. The 41-minute total-recording figure is not a per-clip limit.

7.4 录音与查看记录 / Recording Audio and Reviewing Recordings

取得相关人员同意后，长按记录按键开始录音；录音状态下双击该键停止。保持收音开孔无遮挡，并尽量减少环境噪声。完成后在 App 中查看录音及可用的转写或总结；转写、说话人区分和待办提取可能误判，使用前核对原始录音。传播手册未规定长按秒数和所有录音提示，应按实际反馈确认开始与结束。

After obtaining required consent, press and hold the recording button to start audio recording, then press it twice to stop while recording. Keep microphone openings clear and reduce background noise. Review the recording and available transcript or summary in the app. Transcription, speaker identification, and action extraction can be wrong; check them against the original audio. The source gives no hold duration or complete recording signals; confirm start and stop from actual feedback.

7.5 音乐播放与音量调节 / Music Playback and Volume Control

连接适用音频设备并开始播放后，单击触控板播放或暂停，双击切换下一首，三击切换上一首；前滑提高音量，后滑降低音量。先低音量测试，再调整到舒适水平。手册介绍网易云音乐集成，但美国地区可用性、账号和内容授权应另行确认，不能保证所有音乐服务均支持同一语音操作。

After connecting a suitable audio device and starting playback, tap the touchpad to play or pause, double-tap for the next track, and triple-tap for the previous track. Swipe forward to raise volume and backward to lower it. Test at low volume first. The source describes NetEase Cloud Music integration; confirm U.S. availability, accounts, and content rights separately. Do not assume every music service supports the same voice commands.

7.6 接听与拨打电话 / Receiving and Making Calls

来电时单击触控板接听，通话中双击结束；来电时双击拒接。拨打电话可先在手机上选择联系人并确认眼镜为音频输出设备；如使用语音拨号，应先确认 App 版本、联系人权限和服务支持。行驶或危险环境中不要操作联系人或进行复杂交互。

Tap the touchpad to answer an incoming call, double-tap during a call to end it, or double-tap an incoming call to reject it. To place a call, select a contact on the phone and confirm that the glasses are the audio output. For voice dialing, first check app version, contact permission, and service support. Do not manage contacts or perform complex interactions while driving or in hazardous conditions.

7.7 使用 AI 助手 / Using the AI Assistant

使用前提：眼镜连接正常，所需账号、网络和 AI 服务可用。说出“理想同学”或长按触控板唤醒，再清晰提出问题；双击触控板可退出。视觉问答时保持目标在镜头视野内，注意不采集无关人员或敏感内容。回复为语音或 App 可用页面中的信息，具体呈现按实际版本；本文不展示不存在依据的镜片 AR 界面。

Before use: ensure a connection and the required account, network, and AI service. Say 理想同学 (Lǐxiǎng Tóngxué) or press and hold the touchpad, then ask clearly. Double-tap the touchpad to exit. For visual questions, keep the subject in the camera view and avoid unrelated people or sensitive information. Responses are spoken or shown on available app pages, depending on version; this manual does not depict an unsupported AR lens interface.

深度研究、AI 记忆和录音总结属于辅助功能。重要事实需核查；涉及医疗、法律、金融或驾驶安全时，应使用专业来源而非仅依赖 AI。英文名称不表示产品已支持英语唤醒词或英语服务，请按实际语言设置确认。

Deep research, AI memory, and recording summaries are assistive features. Verify important facts and use professional sources for medical, legal, financial, or driving matters. The English terminology in this manual does not establish an English wake word or English-language service; check actual language settings.

7.8 查看与传输照片、视频 / Viewing and Transferring Photos and Videos

1. 连接眼镜并打开 App 的媒体管理入口，查找新拍摄内容。

1. Connect the glasses and open the media-management entry in the app to find recent captures.

2. 选择需要的照片或视频，按界面导入、保存或传输到手机；允许对应相册或存储权限。

2. Select photos or videos and import, save, or transfer them as prompted; grant the relevant Photos or storage permission.

3. 保持设备靠近并等待完成，再在手机中打开文件确认可用；确认备份后再删除原件。

3. Keep the devices close until transfer completes, then open the files on the phone to verify them. Delete originals only after confirming the backup.

提示：传输方式和速度以当前 App 为准。

NOTE: Transfer method and speed depend on the current app.

7.9 兼容设备与可选功能 / Compatible Devices and Optional Features

手册描述最多同时连接两台设备、保留八台设备绑定关系。手机或电脑切换时检查音频输出和当前活动连接。车辆控制、主驾独立声场及任务联动仅在兼容车辆、适用软件和账号授权条件下使用；手册中的计划发布日期不能代替实际推送确认。美国地区是否支持相关车辆和服务，需另外核实。

The source describes up to two simultaneous connections and eight paired-device relationships. Check audio output and the active connection when switching between a phone and computer. Vehicle controls, a driver-specific audio zone, and task integration require compatible vehicles, software, and account authorization. A planned source release date is not confirmation of delivery. Confirm supported vehicles and services in the United States separately.

8. 充电与电池管理 / Charging and Battery Management

8.1 为眼镜充电 / Charging the Glasses

确保眼镜和盒内接触部位干燥、清洁，自然折叠镜腿并放入盒内。不要弯折镜框或在未就位时强压盒盖。查看实物或 App 的充电状态确认正在补能；若电量不增加，按第 11.2 节排查。手册典型数据为 0–80%约 30 分钟、0–100%约 60 分钟。

Ensure the glasses and case contact areas are dry and clean. Fold the temples naturally and place the glasses inside without bending the frame or forcing the lid. Confirm charging from the actual device or app status. If charge does not increase, see Section 11.2. The source’s typical times are approximately 30 minutes to 80% and 60 minutes to 100% from empty.

8.2 为充电盒进行有线充电 / Wired Charging of the Case

将适用 USB-C 线接入充电盒，并连接符合产品输入要求的电源。放在干燥、通风、稳定的表面，不覆盖设备。手册专门电池章节标示 0–80%约 50 分钟、0–100%约 90 分钟；实际受电源、线缆、温度和电池状态影响。充电输入规格以产品标签为准。

Connect a suitable USB-C cable to the case and a power source matching its input requirements. Place it on a dry, stable, ventilated surface without covering it. The dedicated battery section states approximately 50 minutes to 80% and 90 minutes to 100% from empty. Power source, cable, temperature, and battery condition affect results. Check the product label for charging input specifications.

8.3 为充电盒进行无线充电 / Wireless Charging of the Case

充电盒支持 Qi 无线充电，手册标示最高 15W。将其按无线充电器要求正确放置并对齐充电区域，移除两者之间的金属物或其他异物。确认充电状态后保持位置稳定；无响应时重新对齐或改用有线充电。最高 15W 并不表示必须购买 15W 以上充电板，也不保证所有条件下按最大功率充电。

The case supports Qi wireless charging, stated up to 15 W. Position it according to the charger instructions and align it with the charging area. Remove metal or other foreign objects between the devices. Confirm charging and keep the position stable; realign or use wired charging if there is no response. “Up to 15 W” does not require a charger above 15 W or guarantee maximum power in every condition.

8.4 查看电量与充电状态 / Checking Battery and Charging Status

在 App 眼镜专区或设备页面查看当前可用电量和充电信息；显示项随版本可能不同。不要假设 App 始终显示左右镜腿及充电盒三个独立百分比。状态灯颜色和闪烁含义需按当前正式说明确认，勿只根据未经确认的灯色判断故障。

Check available battery and charging information in the app’s glasses or device page; fields may vary by version. Do not assume three separate percentages for the left temple, right temple, and case. Confirm indicator colors and patterns from current official instructions and do not diagnose faults from unconfirmed colors alone.

8.5 续航影响因素 / Factors Affecting Battery Life

录像、频繁视觉问答、较高音量、反复传输、弱连接和极端温度可能增加耗电。18.8 小时为典型续航描述，不能保证高负载连续运行同样时长。需要长时间外出时，提前检查眼镜和充电盒电量，适当减少不必要的高负载功能。

Video, frequent visual questions, higher volume, repeated transfers, weak connections, and extreme temperatures can increase power use. The typical 18.8-hour figure does not guarantee the same duration under continuous heavy load. Check glasses and case charge before a long outing and reduce unnecessary high-load use.

8.6 长期存放 / Long-Term Storage

长期不使用时，擦净干燥后放在阴凉干燥处，保留适当电量并定期检查眼镜和盒，避免长期亏电。存放电量与检查周期按正式指引，手册未给固定数值。

Clean, dry, and store in a cool, dry place with some charge. Check both glasses and case periodically and avoid prolonged depletion. Follow official storage-charge and interval guidance; no fixed values are specified in the source.

9. 隐私与数据使用提示 / Privacy and Data Use

9.1 拍摄与录音注意事项 / Photography and Recording Considerations

使用前确认场所是否允许拍摄或录音。在医院、学校、办公场所、博物馆、剧场等环境遵守现场规则，避免记录他人证件、屏幕、商业秘密或私人活动。眼镜提示灯不是取得同意的替代方式。录像时隐私灯白色常亮，遮挡提示灯会使拍照或录像无法执行；不要覆盖、改装或规避此提示。

Check whether photography or recording is allowed before use. Follow rules in hospitals, schools, workplaces, museums, and theaters, and avoid capturing identification, screens, trade secrets, or private activities. An indicator is not a substitute for consent. It stays white during video; covering it prevents photos and video. Do not cover, modify, or bypass it.

录音前取得适用法律要求的同意。美国各州及具体情境的录音规则不同，跨州通信也可能涉及多地规则；不要仅因本人参与对话就认为一定可以录音。驾驶、骑行时遵守当地交通规定并避免分心。

Obtain the consent required by applicable law before recording audio. U.S. rules vary by state and circumstances, and interstate conversations may involve more than one jurisdiction. Do not assume your participation alone always permits recording. Follow traffic rules and avoid distraction while driving or cycling.

9.2 应用权限与个人数据 / App Permissions and Personal Data

在手机设置中管理蓝牙、相册、麦克风、通知及必要位置权限，仅授予所需权限。关闭权限可能影响对应功能。账号、记忆、转写等数据的收集、传输、存储和删除按适用隐私政策；不推定仅本地处理或特定服务器所在地。

Manage Bluetooth, Photos, microphone, notifications, and necessary location access in phone settings. Grant only what is needed; disabling access may affect related functions. Applicable privacy policies govern collection, transfer, storage, and deletion for accounts, memory, and transcripts. Do not assume local-only processing or a particular server location.

美国隐私权可能依州法、适用主体和资格条件而不同。若适用，可按官方隐私政策申请访问、更正、删除或退出特定数据处理；本说明书不保证所有用户都具有同一组权利。

U.S. privacy rights depend on state law, the covered entity, and eligibility. Where applicable, use the official privacy-policy process to request access, correction, deletion, or an opt-out from particular processing. This manual does not guarantee identical rights for every user.

9.3 分享与删除内容 / Sharing and Deleting Content

分享照片、视频、录音或 AI 总结前，检查是否包含他人隐私、受保护资料或商业秘密。删除前确认是否已保存所需原件；手机副本、眼镜文件、App 记录和云端数据可能是不同对象，删除其中一处不代表全部删除。公开发布前复核内容和接收对象，不将自动转写或 AI 总结当作未经核查的事实。

Before sharing photos, videos, recordings, or AI summaries, check for privacy, protected material, and trade secrets. Save any needed originals before deletion. Phone copies, glasses files, app records, and cloud data may be separate; deleting one does not establish deletion everywhere. Review the content and recipients before publishing, and do not present unchecked transcripts or AI summaries as verified facts.

9.4 AI 输出的使用限制 / Limitations of AI-Generated Responses

AI 可能因网络、噪声、拍摄角度、识别和语义理解错误而给出不准确或不完整的内容。医疗、法律、金融及安全问题应咨询合适的专业来源。深度研究报告的引用和结论应逐项核实；车辆控制、导航或状态查询，应以车辆实际情况和正式指引为准。AI 记忆或总结也可能遗漏或误记，重要安排应人工确认。

AI may be inaccurate or incomplete because of network conditions, noise, camera angle, recognition, or interpretation errors. Use appropriate professional sources for medical, legal, financial, and safety matters. Verify citations and conclusions in research reports. For vehicle control, navigation, or status, rely on actual conditions and official guidance. Memory and summaries may omit or misremember details; confirm important arrangements yourself.

9.5 转让或停止使用前的准备 / Preparing for Transfer or Discontinued Use

1. 导出需要保留的照片、视频和录音，打开备份确认完整。

1. Export needed photos, videos, and recordings, then open the backups to verify them.

2. 按 App 功能删除设备及账号中的相关个人内容，检查记忆、回忆和云端记录的单独删除入口。

2. Use available app functions to delete relevant device and account content, including separate memory, recollection, or cloud controls.

3. 在 App 解除设备绑定，再按官方指引移除手机等设备上的配对；如提供恢复出厂功能，按该型号正式步骤执行。

3. Unbind in the app and remove phone or other pairings as officially instructed. If a factory-reset function exists, follow the model-specific official steps.

4. 确认设备不再关联个人账号。无法确认删除范围时先联系支持，不编造重置组合键。废弃时交给适用电子产品和电池回收渠道。

4. Confirm the device is no longer linked to your account. Ask support if deletion scope is unclear; do not invent a reset key combination. Use an appropriate electronics and battery recycler for disposal.

10. 清洁、保养与软件更新 / Cleaning, Care, and Software Updates

10.1 镜片与镜架清洁 / Cleaning the Lenses and Frame

将眼镜从电源和充电盒移开。先用清洁、柔软的镜布轻轻去除表面浮尘；颗粒较多时不要用力干擦。镜片污渍按适用蔡司镜片及厂家说明处理，不把镜片护理方式用于整个电子镜架。没有明确清洁指引时，不用流水冲洗、浸泡或向镜架喷清洁液；使用洁净干软布擦拭，并保持收音孔与接口干燥。

Remove the glasses from power and the case. Gently lift loose dust with a clean, soft lens cloth; do not rub hard when grit is present. Follow the applicable ZEISS and manufacturer guidance for lens stains rather than applying lens-care methods to the electronic frame. Without explicit guidance, do not rinse, immerse, or spray liquid onto the frame. Wipe with a clean, dry, soft cloth and keep openings and ports dry.

不要使用粗糙纸巾、衣物、研磨材料、丙酮或强酸强碱清洁剂。含酒精用品是否适用于特定镜片膜层和镜架，应按厂家说明确认，不自行推定。不要拆下镜片、镜腿或转轴清洗。

Do not use rough paper, clothing, abrasives, acetone, or strongly acidic or alkaline cleaners. Check manufacturer guidance before using alcohol-containing products on a specific lens coating or frame. Do not remove lenses, temples, or hinges for cleaning.

10.2 充电接触部位与充电盒保养 / Caring for Charging Contacts and the Case

断开充电线，并将盒从无线充电器取下后再清洁。用干燥软布轻擦可接触的充电部位，避免金属工具刮擦或液体进入盒内。发现腐蚀、变形或难以去除的异物时不要强行处理，应联系支持。重新充电前确认完全干燥；无线充电时检查接触面无金属或异物并正确对齐。

Disconnect the cable and remove the case from the wireless charger before cleaning. Wipe accessible charging areas gently with a dry, soft cloth. Avoid metal tools and liquid inside the case. Contact support for corrosion, deformation, or debris that cannot be removed gently. Ensure complete dryness before charging. Keep wireless charging surfaces free of metal and foreign objects and align them correctly.

10.3 携带与存放 / Carrying and Storage

不佩戴时放入充电盒保护镜片和镜架，先确保产品干燥。不要与钥匙或尖锐物品混放，不要挤压盒盖。避免高温、潮湿、直射阳光和明显粉尘。长时间存放按第 8.6 节检查电量；如曾受液体或强烈冲击，不要仅因外观正常就继续充电。

Use the case to protect lenses and frame when not wearing the glasses, ensuring they are dry first. Keep them away from keys and sharp objects and do not compress the lid. Avoid heat, moisture, direct sun, and heavy dust. Check charge during

extended storage under Section 8.6. After liquid exposure or strong impact, do not resume charging solely because the exterior looks normal.

10.4 检查并安装更新 / Checking for and Installing Updates

1. 连接眼镜和手机，确认电量满足 App 要求且网络稳定。

1. Connect the glasses and phone, meet the app's battery requirement, and ensure a stable network.

2. 在 App 眼镜专区或设备管理页面检查更新，阅读版本说明和兼容条件。

2. Check for updates in the glasses or device page and read release notes and compatibility conditions.

3. 按提示下载和安装；保持连接，不在更新时进行拍摄、解除绑定或人为中断供电。

3. Download and install as prompted. Maintain the connection and avoid capture, unbinding, or deliberately interrupting power during the update.

4. 等待完成，按提示重连或重启，并测试基本拍摄与音频功能。

4. Wait for completion, reconnect or restart if prompted, then test basic capture and audio.

更新推送时间和新增功能以官方实际发布为准。更新失败时记录错误提示，按第 11 章检查连接和网络；反复失败时联系支持，不使用非官方固件或未经证实的强制重置方式。

Actual releases determine update timing and new functions. If an update fails, record the error and check connection and network under Chapter 11. Contact support for repeated failures; do not use unofficial firmware or an unverified forced reset.

11. 故障排查 / Troubleshooting

先检查安全：异常发热、鼓包、异味、进液或明显损坏时直接按第 11.7 节停止使用。普通功能问题按对应表逐项处理，每完成一项都重新测试原问题；避免在没有诊断依据时反复删除账号或恢复出厂。

Check safety first. For unusual heat, swelling, odor, liquid entry, or visible damage, stop under Section 11.7. For ordinary functional issues, work through the appropriate table and retest the original issue after each action. Do not repeatedly delete accounts or factory-reset without a diagnosis.

11.1 无法连接或配对 / Connection or Pairing Issues

检查项目 / Check	处理办法 / Action
眼镜是否有电 / Power	放入充电盒补能，检查可用状态 / Recharge in the case and check availability
蓝牙与权限 / Bluetooth and permissions	开启蓝牙及 App 要求的附近设备权限 / Enable Bluetooth and required Nearby Devices access
距离与干扰 / Distance and interference	靠近手机，远离明显干扰 / Move closer and away from interference
配对状态 / Pairing state	在 App 按当前指引添加，不尝试未核实的盒上键组合 / Follow current Add Device guidance, not assumed case-button combinations
活动连接与账号 / Active connections and account	确认是否达到两台活动连接上限、是否绑定本人账号；不要无故解除全部绑定 / Check the two-active-connection limit and account link; do not erase all pairings unnecessarily
App 版本 / App version	更新 App 后重新打开连接 / Update and reopen the app

复测：确认 App 识别眼镜，再测试音频或照片。仍失败时重启 App 或手机，记录系统版本、设备名称和错误提示后联系支持。

Retest: confirm app recognition and try audio or a photo. If it still fails, restart the app or phone and record system version, device name, and error before requesting support.

11.2 无法充电 / Charging Issues

检查项目 / Check	处理办法 / Action
放置是否正确 / Placement	自然折叠并按盒内形状放好，不强压 / Fold naturally and reseat without force
充电盒是否有电 / Case charge	用适用 USB-C 设备为盒补能 / Charge the case with suitable USB-C equipment
线缆与电源 / Cable and power	检查输入兼容性，尝试状态良好的兼容线缆和电源 / Check compatibility and try an undamaged suitable cable and source
接触部位 / Contacts	完全干燥后用软布清洁；腐蚀或变形时停止 / Clean with a dry soft cloth; stop for corrosion or deformation
无线对齐 / Wireless alignment	移除异物并重新对齐，必要时改用有线 / Remove objects, realign, or use wired charging
环境 / Conditions	移至温和、干燥环境，避免冷热冲击 / Move to moderate, dry conditions without sudden temperature changes

复测：查看 App 或设备实际提示，确认电量有增加。持续无充电或出现异常发热、鼓包、异味时停止使用，不反复试充。

Retest: check actual device or app status and confirm charge increases. Stop for persistent failure, unusual heat, swelling, or odor; do not repeatedly retry charging.

11.3 拍照或录像异常 / Photo or Video Issues

检查项目 / Check	处理办法 / Action
提示灯是否遮挡 / Indicator obstruction	移开遮挡物，不覆盖隐私灯 / Clear the privacy indicator; do not cover it
操作是否正确 / Correct control	记录键单击拍照、双击开始/停止视频 / One button press for a photo; two to start/stop video
电量与存储 / Power and storage	补能，按 App 检查可用空间，先备份再删除 / Recharge, check available space, back up before deletion
画质 / Image quality	轻洁镜头，改善光线并保持头部平稳 / Clean the camera gently, improve lighting, keep steady
单段限制 / Clip limit	核对当前时长设置；手册最高 3 分钟 / Check current duration settings; source states up to 3 min
软件 / Software	确认已完成适用更新 / Confirm applicable updates completed

复测：拍一张照片或一段短视频，在 App 打开验证。持续文件损坏或无法拍摄时记录提示并联系支持。

Retest: capture a photo or short video and open it in the app. Record errors and request support for persistent corruption or capture failure.

11.4 音频或通话异常 / Audio or Call Issues

检查项目 / Check	处理办法 / Action
音量与输出 / Volume and output	低音量起调，确认手机将音频输出至眼镜 / Raise from low volume and select the glasses as phone output
其他设备 / Other devices	检查同时连接设备是否抢占播放；在源设备切换输出 / Check competing playback and switch output on the source
连接 / Connection	靠近音源设备，减少遮挡 / Move closer to the source and reduce obstructions
通话收音 / Call pickup	清除麦克风孔外部遮挡，转到安静环境 / Clear external microphone obstructions and move somewhere quiet
漏音 / Sound leakage	降低音量；敏感通话转用合适的私密方式 / Lower volume or use a suitable private method for sensitive calls

复测：播放短音频或在方便时测试通话，确认双方听音清楚。避免以最大音量测试故障。

Retest with a short audio sample or a convenient test call and confirm intelligibility. Do not test faults at maximum volume.

11.5 AI 助手无法响应 / AI Assistant Not Responding

检查项目 / Check	处理办法 / Action
唤醒方法 / Wake method	说“理想同学”或长按触控板，不用未经支持的英文词 / Say 理想同学 or hold the touchpad; avoid an unsupported English wake word
手机连接 / Phone connection	确认 App 识别眼镜，保持必要后台运行权限 / Confirm app recognition and required background operation
网络与账号 / Network and account	检查联网、登录及地区服务支持 / Check network, sign-in, and regional service support
环境与语言 / Environment and language	减少噪声，清晰说话，核对支持语言 / Reduce noise, speak clearly, check supported language
退出或版本 / Exit state or version	重新唤醒，检查 App 与设备更新 / Wake again and check app/device updates

复测：提出简短普通问题。音频正常但 AI 持续不响应时，区分网络或服务不可用与硬件问题，记录时间和错误后联系支持。

Retest with a short ordinary question. If audio works but AI does not, distinguish network/service availability from hardware faults and record time and errors for support.

11.6 文件同步异常 / File Synchronization Issues

检查项目 / Check	处理办法 / Action
连接 / Connection	保持手机与眼镜靠近且 App 连接正常 / Keep devices close with an app connection
权限 / Permissions	开启导入所需相册或存储权限 / Grant Photos/storage access needed for import
可用空间 / Free space	清理已确认备份的文件，保留足够手机空间 / Remove verified backups as appropriate and free phone space
任务未完成 / Transfer incomplete	等待较大文件传完，不中途断开 / Wait for large files to finish; avoid disconnecting

检查项目 / Check	处理办法 / Action
App 或文件状态 / App or file state	更新 App 并重试单个文件，记录异常提示 / Update the app and retry one file; record errors

复测：先同步一个小文件并打开确认，再传较大文件。未确认备份完整前不要删除眼镜原件；仍失败时联系支持。

Retest with one small file and open it before transferring larger files. Do not delete originals until backup integrity is confirmed. Request support if failure persists.

11.7 何时停止使用并联系支持 / When to Stop Use and Contact Support

出现异常发热、冒烟、鼓包、异味、漏液、镜片破裂、镜架或转轴断裂、接触部位腐蚀、严重进液或强烈冲击后异常时停止使用。若可安全操作，断开外部电源；不强制重启、不私自拆修。普通功能问题排查后仍持续时，也应联系支持并提供第 12.5 节信息。美国地区渠道应先核实，不假设存在专门美国服务点。

Stop use for unusual heat, smoke, swelling, odor, leakage, broken lenses, frame or hinge failure, corroded contacts, major liquid entry, or abnormal behavior after strong impact. Disconnect external power if safe. Do not force a restart or repair it yourself. Persistent ordinary faults also warrant support with the information in Section 12.5. Confirm regional channels rather than assuming a dedicated U.S. service location.

12. 售后服务与保修条款 / After-Sales Service and Warranty Terms

12.1 适用范围 / Scope

本章说明申请服务的准备与流程，不自行设定美国商业保修承诺。传播手册未提供美国保修政策或当地服务网络。申请前确认具体型号、购买渠道和地区资格，并索取适用于该交易的书面保修、退换货和维修条款。

This chapter explains service preparation and process; it does not create a U.S. commercial warranty. The source gives no U.S. warranty policy or local service network. Before requesting service, confirm model, seller, and regional eligibility, and obtain the written warranty, returns, and repair terms applicable to the purchase.

12.2 保修期限与覆盖内容 / Warranty Period and Coverage

保修期限、起算方式、覆盖部件、维修或更换方式，以经核实的购买条款、保修卡和适用法律为准。不得从竞赛资料推定固定年限、免费换新或跨境全国联保。保留订单、购买凭证及随附条款；商业保修不应被理解为取消适用法律赋予消费者的权利。

Confirm duration, start date, covered parts, and repair or replacement remedies from verified purchase terms, warranty documentation, and applicable law. Competition materials do not establish a fixed period, free replacement, or universal cross-border coverage. Keep the order, receipt, and supplied terms. A commercial warranty should not be interpreted as removing applicable legal consumer rights.

12.3 不予保修的情形 / Exclusions

损坏原因可能影响是否属于商业保修，例如事故、误用、液体损坏、未经许可的改装或正常磨损；最终按适用条款、实际检测和法律确定。不要仅因使用第三方配件或维修就断言全部保修失效，需核实是否与故障存在因果关系及适用限制。收费维修前要求说明原因、费用及授权流程。

The cause of damage may affect commercial coverage, including accident, misuse, liquid damage, unauthorized modification, or normal wear. Applicable terms, assessment, and law determine the result. Do not assume all coverage is void solely because of a third-party accessory or repair; check causation and applicable restrictions. Request reasons, a cost estimate, and an authorization process before paid repair.

12.4 申请售后服务的流程 / Requesting Service

1. 通过购买凭证、官方渠道或 App 可用入口确认真实支持渠道，避免将付款信息发给不明客服。

1. Verify support through purchase documentation, official channels, or an available app entry. Do not send payment details to unknown support contacts.

2. 提交型号、购买信息和故障描述，按安全范围内的官方排查指引处理。

2. Provide model, purchase details, and fault description and follow official troubleshooting within safe limits.

3. 寄送前取得受理编号、收件地址、包装要求、费用说明及含锂电池运输指引；不要自行寄送鼓包或漏液设备。

3. Before shipping, obtain a case number, address, packing instructions, costs, and lithium-battery guidance. Do not independently ship swollen or leaking devices.

4. 备份并按第 9.5 节处理个人数据；维修可能清除内容。对受损设备无法删除时告知支持，勿为删数据强行开机。

4. Back up and handle personal data under Section 9.5; service may erase content. If damage prevents deletion, tell support rather than powering a hazardous device to erase it.

5. 检测后核对处理方案并确认授权，收到产品后检查功能和账号状态。

5. Review the assessment and authorize the remedy, then check functions and account state when the product returns.

12.5 申请时需要准备的信息 / Information Required for Service

信息 / Information	准备要点 / What to Prepare
产品身份 / Product identity	产品名、型号、可获取的序列号或设备编号 / Product name, model, available serial or device ID
购买依据 / Purchase evidence	日期、卖家、订单号、凭证和保修文件 / Date, seller, order number, receipt, and warranty documents
故障记录 / Fault record	现象、首次发生时间、频率、错误提示和已试步骤 / Symptoms, first occurrence, frequency, errors, and attempted steps
版本与条件 / Versions and conditions	手机系统、App 和眼镜版本、是否进液或受冲击 / Phone, app, glasses versions, and any liquid/impact exposure
联络与寄送 / Contact and shipping	按渠道要求提供必要联系信息，不公开账号密码 / Required contact details; never disclose passwords

12.6 服务渠道与地区限制 / Service Channels and Regional Limitations

仅使用已核实的厂家、App 或购买渠道公布的服务信息。本稿不提供未确认的美国电话、地址或官网服务承诺。跨境购买时提前确认美国能否维修、是否需寄回原销售地，以及运费、关税和电池运输限制；不要将国内服务或中国境内配镜时效直接外推到美国。

Use only verified manufacturer, app, or seller service information. This draft does not supply an unconfirmed U.S. phone number, address, or website service promise. For cross-border purchases, confirm U.S. repair availability, return to the original sales region, shipping costs, duties, and battery restrictions. Do not extend domestic Chinese service or lens-fulfillment timing to the United States.

附录 A：操作速查表 / Appendix A: Quick Reference Guide

目的 / Task	操作 / Control	检查 / Check
拍照 / Photo	记录键单击 / Press recording button once	镜头和隐私灯无遮挡 / Clear camera and indicator
录像 / Video	记录键双击开始/停止 / Press twice to start/stop	录像白灯常亮；手册单段最高 3 分钟 / Steady white during video; source maximum 3 min per clip
录音 / Audio recording	记录键长按开始；录音时双击停止 / Hold to start; press twice during recording to stop	先取得必要同意 / Obtain required consent
播放暂停 / Play/pause	触控板单击 / Tap touchpad	音频已连接 / Audio connected
切歌 / Tracks	触控板双击下一首、三击上一首 / Double-tap next; triple-tap previous	当前为播放状态 / Playback active
音量 / Volume	触控板前滑提高、后滑降低 / Swipe forward up; backward down	低音量起调 / Start low
通话 / Calls	单击接听；双击结束或拒接 / Tap answer; double-tap end/reject	依来电或通话状态 / Depends on incoming/active state
AI / Assistant	说“理想同学”或长按触控板；双击退出 / Say 理想同学 or hold touchpad; double-tap exit	账号、网络、地区支持 / Account, network, region supported
眼镜充电 / Glasses charging	干燥后自然放入充电盒 / Place dry glasses naturally in case	确认充电状态 / Confirm charging
传输 / Transfer	按 App 媒体管理流程导入 / Import through app media controls	打开备份后再删原件 / Verify backup before deletion

附录 B：中英文术语表 / Appendix B: Bilingual Glossary

中文	English
理想 AI 眼镜 Livis	Li Auto Livis AI Glasses
理想同学 App	Li Auto AI Assistant App (descriptive translation)
理想同学（唤醒词）	理想同学 / Lǐxiǎng Tóngxué (documented wake phrase)
镜架 / 镜腿 / 鼻托	frame / temple / nose pad
记录按键 / 触控板	recording button / touchpad
隐私提示灯	privacy indicator
充电盒 / 充电触点	charging case / charging contacts
第一视角拍摄	first-person capture
开放式音频	open-ear audio
动态照片 / 电子防抖	Live Photo / electronic image stabilization (EIS)
配对 / 绑定 / 同步	pairing / account binding / synchronization
无线升级	over-the-air (OTA) update
视觉问答 / 深度研究	visual questions / deep research
典型续航 / 单段录像时长	typical battery life / per-clip recording duration

中文	English
处方镜片	prescription lenses

附录 C：资料来源 / Appendix C: Sources

页码对应拆分文件 1.pdf 与 2.pdf。查阅日期：2026 年 10 月 12 日。本文未进行独立实机或认证验证。

Pages refer to split files 1.pdf and 2.pdf. Reviewed October 12, 2026; no independent device or certification testing was performed.

资料 / Source	使用范围 / Use
S1. 理想 AI 眼镜 Livis—传播手册，拆分 1.pdf，第 1—10 页 **Livis communication manual, split 1.pdf, pp.1–10**	功能概述、镜架与镜片、尺寸与 IP54、相机、视频、音频。 Features, frame/lenses, dimensions/IP54, camera, video, audio.
S2. 同上，拆分 2.pdf，第 1—14 页 **Same source, split 2.pdf, pp.1–14**	连接、材质、佩戴、电池及充电、AI 与软件、操作速查、App。 Connectivity, materials, fit, battery/charging, AI/software, controls, app.
S3. 原目录、5-8 章.docx、9-12 节.docx **Original outline and supplied Chapters 5–12**	章节结构与原有内容基础，冲突数据以 S1/S2 核对。 Structure and original content; conflicting data checked against S1/S2.

图 3-1、图 4-1 摘自指定传播手册；图 6-1 沿用团队提供的原稿图，须由团队核实其来源和使用权。若采用新增 AI 概念图，应在 AI 使用报告中记录，概念图不作为真实硬件结构依据。

Figures 3-1 and 4-1 are taken from the designated source. Figure 6-1 is retained from the supplied draft; the team should verify its provenance and usage rights. Record any added AI concept illustration in the AI-use report; it is not a hardware reference.